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A Three-Interface Data-based Wireless Communication Timeout Fault Diagnosis Method for China Train Control System

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Abstract—**The high-speed train control system is a crucial core equipment ensuring punctual departure and efficient operation of high-speed trains, while wireless timeout faults have always been a challenging issue in its operation and maintenance. Currently, fault diagnosis and maintenance heavily rely on manual planned inspections and post-event handling, leading to low diagnostic efficiency. To address this problem, this paper proposes an intelligent diagnosis method based on three-interface data, utilizing the random forest classification model to improve diagnostic efficiency and reduce the probability of fault occurrences. The paper introduces the communication principles and typical case analysis between the train vehicle speed protection equipment and the ground radio block center in the high-speed train control system. It then elaborates on the fault diagnosis method based on three-interface data, aiming to provide an effective means for dynamic health management of high-speed train control system equipment, ensuring safe and efficient operation of high-speed trains. Through the research presented in this paper, strong support can be provided for handling wireless timeout faults in high-speed train control systems, contributing to the improvement of train operation quality and efficiency.**

Keywords-train control system; fault diagnosis; Three-interface data; random forest

I. INTRODUCTION

The high-speed train control system is one of the crucial core components for ensuring the on-time departure and efficient operation of high-speed trains. According to the design specifications of China's high-speed railways, the high-speed railways with a design speed of 250 kilometers per hour or above adopt the CTCS-3 train control system. In the CTCS-3 train control system, the wireless communication between the vehicle Automatic Train Protection (ATP) equipment and the Ground Radio Block Center (RBC) is achieved through the GSM-R network, enabling bidirectional transmission of messages[1]. This facilitates the Train Control Center to issue mobile authorization commands to high-speed trains and ensures the enforcement of safe operating intervals. Since the first operation of the Wuhan-Guangzhou high-speed railway, which utilized the CTCS-3 train control system in 2009, the wireless communication timeout issue between the train's vehicle CTCS-3 equipment and the ground control system has consistently posed a significant challenge in the maintenance and improvement of railway signal reliability. As existing and newly constructed railway lines are operating at speeds of up to 350 kilometers per hour, this communication timeout issue has gradually become a critical factor restricting further speed increases and affecting train operation efficiency. On the other hand, in current practical production, the diagnosis and maintenance of the train-ground wireless communication timeout faults still heavily rely on manual planned inspections and post-event repairs. This traditional approach results in low diagnostic efficiency and lacks efficient intelligent diagnostic methods.

Based on the background mentioned above, this article proposes an intelligent diagnostic method for train-ground wireless communication timeout faults based on three-interface data. The core of this method involves utilizing a Random Forest classification model to achieve fault diagnosis. Firstly, the data from the three interfaces (A , Abis , and PRI) of the high-speed train control system is subjected to preprocessing and divided into training and testing sets[2]. Next, the training set data is used to train the Random Forest classifier, resulting in a diagnostic model for train-ground wireless communication faults. Subsequently, the diagnostic model is utilized to detect and classify faults in the testing set data. Finally, an evaluation and analysis of metrics, including accuracy, are conducted on the diagnostic model. The results indicate that the proposed method efficiently diagnoses train-ground wireless communication timeout faults with a high level of accuracy [3].

II. VEHICLE-GROUND WIRELESS COMMUNICATION PRINCIPLE OF THE HIGH-SPEED TRAIN CONTROL SYSTEM

The CTCS-3 train control system utilizes wireless communication (GSM-R) to establish continuous, bidirectional, and high-capacity message transmission between trains and the ground control center. It also employs track circuits to check the occupancy status of train block sections and utilizes ground transponder to calibrate positioning information. In the CTCS-3 train control system, the rail vehicle equipment(Fig.1) is responsible for receiving ground data and command information, generating speed mode curves, monitoring train operation, and ensuring the safety of train operations. The ground signaling equipment(Fig.2) generates and sends movement authorities to trains based on information received from interlocking devices, adjacent RBC, temporary speed restriction servers, and interactions with vehicle train equipment. This allows trains to safely operate within the jurisdiction of the RBC, ensuring train interval control and train protection on the designated railway tracks. The CTCS-3 train-ground wireless communication transmission system consists of vehicle wireless transmission equipment, the Ground Radio Block Center (RBC), and the GSM-R network, enabling wireless communication between trains and the ground. If any side experiences abnormalities, it may result in train-ground wireless communication timeout faults. The schematic diagram of the train-ground wireless communication transmission system is shown in Fig.3.

Fig. 1: Rail vehicle equipment.

Fig. 2: The ground signaling equipment.

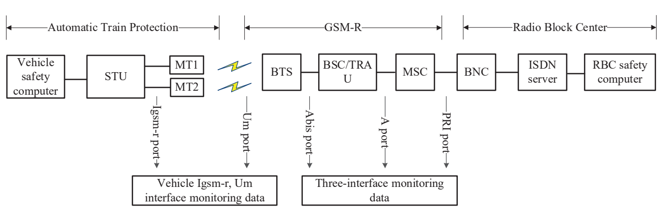

Fig. 3: Structure of the train-ground wireless transmission system.

In the transmission process, the communication data between the CTCS-3 train control system and the ground passes through numerous network elements. The interfaces involved primarily include Igsm-r, Um, Abis, A, and PRI. The monitoring data from the Igsm-r and Um interfaces can only be collected by trains equipped with air interface devices and is not universally applicable. Therefore, this study primarily relies on existing PRI, A, and Abis interface monitoring data, including the Abis-interface data between the BTS (Base Transceiver Station) and BSC (Base Station Controller). The main focus of monitoring is on handover between cells, signal interactions, uplink and downlink reception levels, and quality of vehicle terminal modules, reflecting the quality of communication network utilization. Additionally, A-interface signaling messages and handover events between MSC and BSC/TRAU are also included in the study. The PRI-interface data exchanged between the MSC (Mobile Switching Center) and RBC (Radio Block Center) includes both wireless communication aspects (physical layer, data link layer, network layer, and transport layer) and signaling-related content (security layer, application layer).

III. Typical Case Analysis

In practical applications, various factors such as unstable ATP equipment operation, RBC or ATP software defects, communication G-network device obstacles, external radio frequency interference, and other reasons can lead to train-ground wireless communication timeout faults. In some cases, these issues can even result in train downgrading, significantly impacting the efficiency of train operations[4].

A. The Vehicle Equipment's MT Radio Fault

The characteristic manifestation of such a fault is that wireless connection timeouts occur at each RBC handover zone. Let's further analyze the data to gain more insights.

On November 13, 2022, during the operation of the CRH380D-1546 high-speed train on the G7615 service from Nanjing South Station, multiple wireless timeout brake failure alarms occurred. Analysis of the data from the three interfaces (PRI port call records or A port call records) revealed that only one vehicle MT (Mobile Terminal) was operational, as shown in Fig. 4 and 5.

序号	触发时间	MSISDN/IMSI	机车号	车次号	线路	消息类型	BSSMAP类型	SCCP类型
13	2022-11-13 17:35:18.7010	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
14	2022-11-13 17:36:12.8420	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
15	2022-11-13 17:37:55.4760	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
16	2022-11-13 17:43:19.1460	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
17	2022-11-13 17:44:42.5340	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
18	2022-11-13 17:45:19.7640	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
19	2022-11-13 17:45:58.9480	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
20	2022-11-13 17:46:49.0930	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
21	2022-11-13 17:47:35.3600	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
22	2022-11-13 17:48:09.3110	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
23	2022-11-13 17:48:34.2810	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
24	2022-11-13 17:49:21.1060	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
25	2022-11-13 17:49:52.4580	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
26	2022-11-13 17:50:38.8010	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
27	2022-11-13 17:51:09.2770	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
28	2022-11-13 17:51:36.8490	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
29	2022-11-13 17:52:29.9380	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1
30	2022-11-13 17:53:12.7990	14984184065	115461	G7615	宁杭	BSSMAP	HANDOVER PERFORMED	DT1

Fig. 4: A interface call records for service G7615.

序号	触发时间	MSISDN/IMSI	RBC号码	机车号	车次号	线路	Layer	消息类型	子类型
4	2022-11-13 17:33:46.577	14984184065	92410502	115461	G7615	宁杭	Physical	Signaling	CONNECT ACK
5	2022-11-13 17:33:48.280	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	SABME
6	2022-11-13 17:33:48.374	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	UA
7	2022-11-13 17:33:49.108	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	I
8	2022-11-13 17:33:49.108	14984184065	92410502	115461	G7615	宁杭	Net	NPDU	Data
9	2022-11-13 17:33:49.139	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	I
10	2022-11-13 17:33:49.139	14984184065	92410502	115461	G7615	宁杭	Net	NPDU	Data
11	2022-11-13 17:33:49.139	14984184065	92410502	115461	G7615	宁杭	Transport	TPDU	CR
12	2022-11-13 17:33:49.139	14984184065	92410502	115461	G7615	宁杭	Safety	SAPDU	AU1
13	2022-11-13 17:33:49.233	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	RR
14	2022-11-13 17:33:49.249	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	RR
15	2022-11-13 17:33:49.827	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	I
16	2022-11-13 17:33:49.827	14984184065	92410502	115461	G7615	宁杭	Net	NPDU	Data
17	2022-11-13 17:33:49.874	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	I
18	2022-11-13 17:33:49.874	14984184065	92410502	115461	G7615	宁杭	Net	NPDU	Data
19	2022-11-13 17:33:49.874	14984184065	92410502	115461	G7615	宁杭	Transport	TPDU	CC
20	2022-11-13 17:33:49.874	14984184065	92410502	115461	G7615	宁杭	Safety	SAPDU	AU2
21	2022-11-13 17:33:50.452	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	RR
22	2022-11-13 17:33:50.467	14984184065	92410502	115461	G7615	宁杭	DL	HDLC frame	RR
23	2022-11-13 17:33:50.717	14984184065	92410502	115461	G7615	宁杭	DL		

Fig. 5: PRI interface call records for service G7615.

Based on the analysis of the three-interface data mentioned above, it can be observed that if the vehicle equipment operates with a single radio (MT), i.e., one MT radio malfunctioning, there are clear characteristics in the data records. At this point, it can serve as a signal to prompt vehicle equipment maintenance personnel to repair or replace the faulty radio.

B. *Wireless Communication Interference*

The characteristics of network interference are characterized by suddenness and repetitive failures. When the high-speed train passes through an interference zone during operation, it experiences sudden wireless connection timeouts. However, once the train exits the interference zone, the wireless communication automatically returns to normal. Subsequently, other trains passing through the same location also encounter similar faults.

During the analysis of wireless connection timeouts, it was observed that certain fixed locations frequently experienced such issues. After conducting on-site inspections and tests, it was discovered that there were telecommunication operator base stations nearby or interfering signals present. Once the interference was eliminated, the fault rate significantly decreased. In such operational scenarios, the key focus of analysis and investigation should be on GSM-R network interference. Below is a specific case analysis.

On December 1, 2022, during the operation of the CR380D-1584 high-speed train on the G2333 service, a wireless connection timeout fault alarm occurred at the location of 254KM+805M on the Hangzhou-Changsha Passenger Dedicated Line. Based on the Abis interface data from the three interfaces, the GSM-R network measurement report is shown in Fig. 6. At 9:38:26.490, the uplink communication quality value changed to 7 (measuring the level of signal quality, ranging from 0 to 7, where 0 indicates the best quality and 7 indicates the worst quality). Simultaneously, the downlink signal level and communication quality measurement values were missing.

序号	触发时间	MSISDN/IMSI	机车号	车次号	线路	小区	LAC	CI	公里标	当前速度	ARFCN	MR序号	上行电平值	上行通信质量	下行电平值	下行通信质
784	2022-12-01 09:38:23.1470	14986173856	115841	G2333	杭长	ZJ-TW09	16665	22	K252+999	300	1005	53	-59	0	-56	
785	2022-12-01 09:38:23.6170	14986173856	115841	G2333	杭长	ZJ-TW09	16665	22	K252+999	300	1005	54	-58	0	-56	
786	2022-12-01 09:38:24.1080	14986173856	115841	G2333	杭长	ZJ-TW09	16665	22	K252+999	300	1005	55	-59	0	-55	
787	2022-12-01 09:38:24.5770	14986173856	115841	G2333	杭长	ZJ-TW09	16665	22	K252+999	300	1005	56	-58	0	-57	
788	2022-12-01 09:38:25.0690	14986173856	115841	G2333	杭长	ZJ-TW09	16665	22	K252+999	300	1005	57	-58	0	-57	
789	2022-12-01 09:38:25.5890	14986173856	115841	G2333	杭长	ZJ-TW09	16665	22	K252+999	300	1005	58	-59	0		
790	2022-12-01 09:38:26.4940	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	0	-104	7		
791	2022-12-01 09:38:26.9620	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	1	-47	7		
792	2022-12-01 09:38:27.4550	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	2	-47	7		
793	2022-12-01 09:38:27.9230	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	3	-47	7		
794	2022-12-01 09:38:28.4170	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	4	-47	7		
795	2022-12-01 09:38:28.8840	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	5	-47	7		
796	2022-12-01 09:38:29.3760	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	6	-47	7		
797	2022-12-01 09:38:29.8450	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	7	-47	7		
798	2022-12-01 09:38:30.3370	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	8	-47	7		
799	2022-12-01 09:38:30.8070	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	9	-47	7		
800	2022-12-01 09:38:31.2980	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	10	-47	7		
801	2022-12-01 09:38:31.7670	14986173856	115841	G2333	杭长	ZJ-TW10	16665	23	K252+999	300	1003	11	-47	7		

Fig. 6: Abis interface data for service

Based on this inference, when the wireless connection timeout fault alarm occurs, there may be interference in the communication at that location. If the wireless connection timeout fault continues to occur at that location, it is advisable to notify communication equipment maintenance personnel to conduct network quality measurements on-site. By identifying the specific faulty components, locate the faulty component and replace it.

C. *RBC Device Failur*

Once the RBC (Radio Block Center) equipment malfunctions, all trains running within its jurisdiction will experience wireless connection timeout fault alarms. The specific reasons for such alarms may be due to interruptions in communication between the RBC and the GSM-R network or self-restart of the RBC system. A specific case analysis is provided below.

On April 5, 2023, at 6:53:48, both the G7554 and G2668 train services experienced wireless connection timeout fault alarms while operating on the Hangzhou-Changsha high-speed line. The PRI interface data from the three interfaces is shown in Fig. 7 and 8. It can be observed that the data exchange between the ATP (Automatic Train Protection) equipment and RBC devices for these two trains encountered delays of over 20 seconds without a response. This situation may indicate a malfunction in the RBC equipment. In such cases, it is advisable to prompt RBC equipment maintenance personnel to inspect the operational status of the equipment and consider replacing any faulty components.

序	触发时间	MSISDN/IMSI	RBC号码	机车号	车次号	线路	Layer	子类型	Direction	Explain
291	2023-04-05 06:52:55.870	14984773948	92410901	335172	G7554	杭黄	Application	M24 General message	RBC->OBU	RBC—>ATP: NID_MESSAGE = 2
293	2023-04-05 06:52:59.213	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE = 1
294	2023-04-05 06:52:59.510	14984773948	92410901	335172	G7554	杭黄	Application	M24 General message	RBC->OBU	RBC—>ATP: NID_MESSAGE = 2
295	2023-04-05 06:53:05.213	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE = 1
296	2023-04-05 06:53:05.744	14984773948	92410901	335172	G7554	杭黄	Application	M24 General message	RBC->OBU	RBC—>ATP: NID_MESSAGE = 2
297	2023-04-05 06:53:11.213	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE = 1
298	2023-04-05 06:53:11.463	14984773948	92410901	335172	G7554	杭黄	Application	M24 General message	RBC->OBU	RBC—>ATP: NID_MESSAGE = 2
300	2023-04-05 06:53:17.212	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE = 1
301	2023-04-05 06:53:17.697	14984773948	92410901	335172	G7554	杭黄	Application	M24 General message	RBC->OBU	RBC—>ATP: NID_MESSAGE = 2
302	2023-04-05 06:53:23.212	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE = 1
303	2023-04-05 06:53:23.431	14984773948	92410901	335172	G7554	杭黄	Application	M24 General message	RBC->OBU	RBC—>ATP: NID_MESSAGE =
304	2023-04-05 06:53:25.400	14984773948	92410901	335172	G7554	杭黄	Application	M132 MA Request	OBU->RBC	ATP—>RBC: NID_MESSAGE =
305	2023-04-05 06:53:25.696	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE =
306	2023-04-05 06:53:31.712	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE =
309	2023-04-05 06:53:36.711	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE =
310	2023-04-05 06:53:42.680	14984773948	92410901	335172	G7554	杭黄	Application	M136 Train Position Report	OBU->RBC	ATP—>RBC: NID_MESSAGE = 1

Fig. 7: PRI interface data for service G7554.

序号	触发时间	MSISDN/IMSI	RBC号码	机车号	车次号	线路	Layer	子类型	Directi	RBCID	公里标	Explain
4610	2023-04-05 06:53:13.197	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K154+452	ATP—>RBC: NID_MESSA
4620	2023-04-05 06:53:13.525	14984073700	92410901	1120871	G2668	杭黄	Application	M24 General m	RBC->OBU	11468801	K154+452	RBC—>ATP: NID_MESSA
4632	2023-04-05 06:53:19.212	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K154+311	ATP—>RBC: NID_MESSA
4642	2023-04-05 06:53:19.790	14984073700	92410901	1120871	G2668	杭黄	Application	M24 General m	RBC->OBU	11468801	K154+311	RBC—>ATP: NID_MESSA
4650	2023-04-05 06:53:20.467	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K154+288	ATP—>RBC: NID_MESSA
4664	2023-04-05 06:53:20.[illegible]	14984073700	92410901	1120871	G2668	杭黄	Application	M24 General m	RBC->OBU	11468801	K154+288	RBC—>ATP: NID_MESSA
4676	2023-04-05 06:53:26.462	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K154+158	ATP—>RBC: NID_MESS
4689	2023-04-05 06:53:30.946	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K154+068	ATP—>RBC: NID_MESS
4702	2023-04-05 06:53:36.946	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K153+944	ATP—>RBC: NID_MESS
4715	2023-04-05 06:53:41.180	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K153+854	ATP—>RBC: NID_MESS
4728	2023-04-05 06:53:42.961	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K153+821	ATP—>RBC: NID_MESS
4741	2023-04-05 06:53:46.430	14984073700	92410901	1120871	G2668	杭黄	Application	M132 MA Reque	OBU->RBC	11468801	K153+751	ATP—>RBC: NID_MESS
4754	2023-04-05 06:53:48.961	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K153+710	ATP—>RBC: NID_MESS
4767	2023-04-05 06:53:54.961	14984073700	92410901	1120871	G2668	杭黄	Application	M136 Train Po	OBU->RBC	11468801	K153+628	ATP—>RBC: NID_MESS
4777	2023-04-05 06:53:55.648	14984073700	92410901	1120871	G2668	杭黄	Application	M156 Terminat	OBU->RBC	11468801	K153+628	ATP—>RBC: NID_MESS
4790	2023-04-05 06:54:10.897	14984073700	92410901	1120871	G2668	杭黄	Application	M156 Terminat	OBU->RBC	11468801	K153+628	ATP—>RBC: NID_MESS

Fig. 8: PRI interface data for service G2668.

IV. Fault Diagnosis Method Based on Three-Interface Data

This paper proposes a method for diagnosing wireless connection timeout faults in high-speed train control systems based on the analysis of three-interface data using a random forest algorithm model[5]. The specific steps of the method include the following steps.

A. *Data Preprocessing for Three-Interface Data*

In the random forest algorithm, the input values must be integers or floating-point numbers. Therefore, the first step is to merge the data from the A, Abis, and PRI ports into a single dataset. Next, the input data undergoes preprocessing, where strings are converted into integers or floating-point numbers. The process involves filling in missing data, removing noise and irrelevant information, and extracting key terms and phrases.

B. *Data Partitioning*

The preprocessed data is divided into training data and testing data. The training set will be used for training the subsequent decision trees, while the testing set will be used to validate the performance of the decision trees. In total, there are 4770 data points, and after merging the A, Abis, and PRI port data, there are 1590 data points. Out of these, 1200 data points are selected as the training dataset, and the remaining 390 data points are used as the testing dataset. List the fields that affect the fault diagnosis result, including trigger time, MSISDN/IMSI, train number, cell, uplink and downlink communication quality, message type, subtype, Direction, Explian, and the corresponding parameter values and precision.

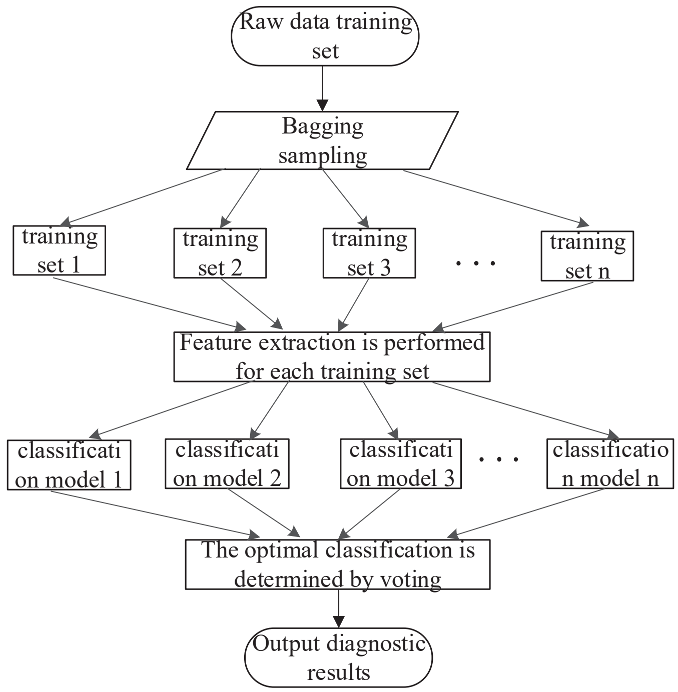

Fig. 9: Flow chart of random forest algorithm.

C. Fault Classification Model Training

The Bagging method is used for training the model. In this approach, samples are randomly selected with replacement from the training set, and decision trees are trained using randomly selected subsets of features. This setting increases the independence of decision tree construction, enhancing the model's resilience to noise and improving its generalization ability. The Bagging method is repeatedly applied to the training set until all the required decision trees are constructed. The final output of the random forest algorithm is determined by aggregating the results from all the decision trees, the specific process is shown in Fig. 9.

V. Conclusion and Future Work

The paper proposes an intelligent fault diagnosis method for wireless communication timeout in high-speed train control systems based on three-interface data. This method utilizes the random forest classification model to accomplish fault diagnosis.

In the next step, the authors plan to leverage Prognostics and Health Management (PHM) technology for predictive analysis of wireless communication timeout faults. This approach aims to achieve dynamic health management of the control system equipment, further reducing the probability of wireless communication timeout faults occurring between trains and the ground control center.

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